



GNU nano 8.7.1

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <string.h>

int main() {
    char command[100];

    printf("Enter a Linux command: ");
    scanf("%s", command);

    pid_t pid = fork();

    if (pid < 0) {
        printf("Fork failed!\n");
        return 1;
    }

    else if (pid == 0) {
        printf("Child Process\n");
        printf("Child PID : %d\n", getpid());

        printf("Executing command: %s\n", command);

        execlp(command, command, NULL);

        printf("Command execution failed!\n");
    }

    else {
        printf("Parent Process\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        wait(NULL);
    }
}
```



keerthi_priya@LAPTOP-OSGV



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```
int main() {
    char command[100];

    printf("Enter a Linux command: ");
    scanf("%s", command);

    pid_t pid = fork();

    if (pid < 0) {
        printf("Fork failed!\n");
        return 1;
    }

    else if (pid == 0) {
        printf("Child Process\n");
        printf("Child PID : %d\n", getpid());

        printf("Executing command: %s\n", command);

        execlp(command, command, NULL);

        printf("Command execution failed!\n");
    }

    else {
        printf("Parent Process\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        wait(NULL);

        printf("Child process completed.\n");
    }

    return 0;
}
```

```
keerthi_priya@LAPTOP-OSGVCE7A:~$ mkdir OS_LAB
keerthi_priya@LAPTOP-OSGVCE7A:~$ cd OS_LAB
keerthi_priya@LAPTOP-OSGVCE7A:~/OS_LAB$ nano task1.c
keerthi_priya@LAPTOP-OSGVCE7A:~/OS_LAB$ gcc task1.c -o task1
keerthi_priya@LAPTOP-OSGVCE7A:~/OS_LAB$ ./task1
Enter a Linux command: ls
Parent Process
Parent PID : 641
Child PID : 642
Child Process
Child PID : 642
Executing command: ls
task1 task1.c
Child process completed.
keerthi_priya@LAPTOP-OSGVCE7A:~/OS_LAB$ |
```

Child process completed.

```
keerthi_priya@LAPTOP-OSGVCE7A:~/OS_LAB$ uname -a
```

```
Linux LAPTOP-OSGVCE7A 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
```

```
keerthi_priya@LAPTOP-OSGVCE7A:~/OS_LAB$ lscpu
```

```
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          46 bits physical, 48 bits virtual
Byte Order:            Little Endian
CPU(s):                18
On-line CPU(s) list:   0-17
Vendor ID:             GenuineIntel
Model name:            Intel(R) Core(TM) Ultra 5 125H
CPU family:            6
Model:                170
Thread(s) per core:    2
Core(s) per socket:    9
Socket(s):             1
Stepping:              4
BogoMIPS:              5990.40
Flags:                 fpu vme de pse tsc msr pae mce cx8 apic sep mtr
r pge mca cmov pat pse36 clflush mmx fxsr sse s
se2 ss ht syscall nx pdpe1gb rdtscp lm constant
_tsc rep_good nopl xtopology tsc_reliable nonst
op_tsc cpuid tsc_known_freq pni pclmulqdq vmx s
sse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe p
opcnt tsc_deadline_timer aes xsave avx f16c rdr
and hypervisor lahf_lm abm 3dnowprefetch ssbd i
brs ibpb stibp ibrs_enhanced tpr_shadow ept vpi
d ept_ad fsgsbase tsc_adjust bmil avx2 smep bmi
2 erms invpcid rdseed adx smap clflushopt clwb
sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni
vnni umip waitpkg gfni vaes vpclmulqdq rdpid mo
vdiri movdir64b fsrm md_clear serialize ibt flu
sh_lld arch_capabilities
```

Virtualization features:

```
Virtualization:      VT-x
Hypervisor vendor:   Microsoft
Virtualization type: full
```

Caches (sum of all):

```
L1d:                432 KiB (9 instances)
L1i:                576 KiB (9 instances)
```

```

L1i:          576 KiB (9 instances)
L2:          18 MiB (9 instances)
L3:          18 MiB (1 instance)
NUMA:
  NUMA node(s):      1
  NUMA node0 CPU(s): 0-17
Vulnerabilities:
  Gather data sampling: Not affected
  Ghostwrite:          Not affected
  Indirect target selection: Not affected
  Itlb multihit:       Not affected
  L1tf:                Not affected
  Mds:                 Not affected
  Meltdown:            Not affected
  Mmio stale data:     Not affected
  Old microcode:       Not affected
  Reg file data sampling: Not affected
  Retbleed:            Mitigation; Enhanced IBRS
  Spec rstack overflow: Not affected
  Spec store bypass:   Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1:          Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2:          Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI BHI_DIS_S
  Srbds:               Not affected
  Tsa:                 Not affected
  Tsx async abort:     Not affected
  Vmscape:              Not affected

```

```

keerthi_priya@LAPTOP-OSGVCE7A:~/OS_LAB$ ps

```

```

  PID TTY          TIME CMD
  346 pts/0    00:00:00 bash
  674 pts/0    00:00:00 ps

```

```

keerthi_priya@LAPTOP-OSGVCE7A:~/OS_LAB$ ps -ef

```

```

UID          PID    PPID  C STIME TTY          TIME CMD
root           1        0  0 16:08 ?           00:00:01 /sbin/init
root           2        1  0 16:08 hvc0       00:00:00 /init
root           6        2  0 16:08 hvc0       00:00:00 plan9 --control-socket 7
root          46        1  0 16:08 ?           00:00:00 /usr/lib/systemd/systemd
systemd+      81        1  0 16:08 ?           00:00:00 /usr/lib/systemd/systemd

```

```
keerthi_priya@LAPTOP-OSGVCE7A:~/OS_LAB$ ps -ef
```

UID	PID	PPID	C	STIME	TTY	TIME	CMD
root	1	0	0	16:08	?	00:00:01	/sbin/init
root	2	1	0	16:08	hvc0	00:00:00	/init
root	6	2	0	16:08	hvc0	00:00:00	plan9 --control-socket 7
root	46	1	0	16:08	?	00:00:00	/usr/lib/systemd/systemd
systemd+	81	1	0	16:08	?	00:00:00	/usr/lib/systemd/systemd
root	92	1	0	16:08	?	00:00:00	/usr/lib/systemd/systemd
root	133	1	0	16:08	?	00:00:00	/bin/sh /usr/lib/systemd
root	138	1	0	16:08	?	00:00:00	/usr/sbin/cron -f -P
message+	143	1	0	16:08	?	00:00:00	@dbus-daemon --system --
root	168	1	0	16:08	?	00:00:00	/usr/bin/python3 /usr/bi
root	172	1	0	16:08	?	00:00:00	/usr/lib/systemd/systemd
syslog	192	1	0	16:08	?	00:00:00	/usr/sbin/rsyslogd -n -i
root	251	1	0	16:08	tty1	00:00:00	/usr/sbin/agetty --nores
_chrony	272	133	0	16:08	?	00:00:00	/usr/sbin/chronyd -n -F
_chrony	283	272	0	16:08	?	00:00:00	/usr/sbin/chronyd -n -F
root	293	1	0	16:08	?	00:00:00	/usr/bin/python3 /usr/sh
root	344	2	0	16:08	?	00:00:00	/init
root	345	344	0	16:08	?	00:00:00	/init
keerthi+	346	345	0	16:08	pts/0	00:00:00	-bash
root	347	2	0	16:08	?	00:00:00	login -- keerthi_priya
keerthi+	389	1	0	16:08	?	00:00:00	/usr/lib/systemd/systemd
keerthi+	393	389	0	16:08	?	00:00:00	(sd-pam)
keerthi+	419	347	0	16:08	pts/1	00:00:00	-bash
keerthi+	684	346	0	16:26	pts/0	00:00:00	ps -ef

```
keerthi_priya@LAPTOP-OSGVCE7A:~/OS_LAB$ top
```

```
top - 16:27:57 up 19 min, 1 user, load average: 0.02, 0.03, 0.00
Tasks: 24 total, 1 running, 23 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7651.1 total, 6970.3 free, 598.1 used, 234.2 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7053.1 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	24084	15360	11580	S	0.0	0.2	0:01.02	systemd
2	root	20	0	3180	2208	2072	S	0.0	0.0	0:00.03	init-systemd(Ub
6	root	20	0	3180	2048	1940	S	0.0	0.0	0:00.00	init
46	root	19	-1	50424	16828	15556	S	0.0	0.2	0:00.23	systemd-journal
81	systemd+	20	0	22416	14520	12028	S	0.0	0.2	0:00.06	systemd-resolve
92	root	20	0	35440	12404	9236	S	0.0	0.2	0:00.45	systemd-udev
133	root	20	0	2888	1948	1820	S	0.0	0.0	0:00.11	chronyd-starter

```

keerthi+ 389 1 0 16:08 ? 00:00:00 /usr/lib/systemd/systemd
keerthi+ 393 389 0 16:08 ? 00:00:00 (sd-pam)
keerthi+ 419 347 0 16:08 pts/1 00:00:00 -bash
keerthi+ 684 346 0 16:26 pts/0 00:00:00 ps -ef

```

```

keerthi_priya@LAPTOP-OSGVCE7A:~/OS_LAB$ top

```

```

top - 16:28:09 up 20 min, 1 user, load average: 0.01, 0.03, 0.00
Tasks: 24 total, 1 running, 23 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.1 sy, 0.0 ni, 99.9 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7651.1 total, 6969.4 free, 598.9 used, 234.2 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 7052.2 avail Mem

```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	24084	15360	11580	S	0.0	0.2	0:01.02	systemd
2	root	20	0	3180	2208	2072	S	0.0	0.0	0:00.03	init-systemd(Ub
6	root	20	0	3180	2048	1940	S	0.0	0.0	0:00.00	init
46	root	19	-1	50424	16828	15556	S	0.0	0.2	0:00.23	systemd-journal
81	systemd+	20	0	22416	14520	12028	S	0.0	0.2	0:00.06	systemd-resolve
92	root	20	0	35440	12404	9236	S	0.0	0.2	0:00.45	systemd-udev
133	root	20	0	2888	1948	1820	S	0.0	0.0	0:00.11	chronyd-starter
138	root	20	0	4472	3128	2880	S	0.0	0.0	0:00.01	cron
143	message+	20	0	8788	5340	4664	S	0.0	0.1	0:00.06	dbus-daemon
168	root	20	0	44096	29780	14636	S	0.0	0.4	0:00.31	networkd-dispat
172	root	20	0	18432	9384	8148	S	0.0	0.1	0:00.17	systemd-logind
192	syslog	20	0	220548	5864	4624	S	0.0	0.1	0:00.16	rsyslogd
251	root	20	0	5492	2712	2524	S	0.0	0.0	0:00.02	agetty
272	_chrony	20	0	23564	11196	7220	S	0.0	0.1	0:00.14	chronyd
283	_chrony	20	0	12096	2340	1696	S	0.0	0.0	0:00.01	chronyd
293	root	20	0	123460	32708	18100	S	0.0	0.4	0:00.25	unattended-upgr
344	root	20	0	3184	1112	980	S	0.0	0.0	0:00.00	SessionLeader
345	root	20	0	3200	1252	1108	S	0.0	0.0	0:00.34	Relay(346)
346	keerthi+	20	0	6268	5600	3928	S	0.0	0.1	0:00.09	bash
347	root	20	0	8648	5536	4692	S	0.0	0.1	0:00.01	login
389	keerthi+	20	0	22340	13624	11000	S	0.0	0.2	0:00.16	systemd
393	keerthi+	20	0	22916	4088	2088	S	0.0	0.1	0:00.00	(sd-pam)
419	keerthi+	20	0	6136	5416	3864	S	0.0	0.1	0:00.02	bash
694	keerthi+	20	0	8172	6008	3900	R	0.0	0.1	0:00.05	top